

First year Aerospace Engineering

Mathematics Recap - Differentials

Marcos Carrera Valle

Simple Derivatives - 6 points

Find the derivatives of the following functions

1) $f(x) = 5x^3 - 3x^2 - x + 4$

2) $f(x) = \sin(x)$

3) $f(x) = \cos(x)$

4) $f(x) = \tan(x)$

5) $f(x) = e^x$

6) $f(x) = \ln x$

Complex derivatives - 8 points

Find the derivatives of the following functions

1) $f(x) = 2xe^x$

2) $f(x) = \frac{e^x}{x}$

3) $f(x) = \sin(3x^2 + 2)$

4) $f(x) = a^x$

Complex derivatives - 5 points

For the function $f(x) = 2x^3 + 3x^2 - 12x$, $-3 < x < 2$

Find

- 1) The slope at the origin - 1 point
- 2) The absolute maxima - 2 points
- 3) The absolute minima - 2 points

Answers

Slide 1:

$$f'(x) = 15x^2 - 6x - 1, f'(x) = \cos(x), f'(x) = -\sin(x), f'(x) = \sec^2(x), f'(x) = e^x, f'(x) = \frac{1}{x}$$

Slide 2: $f'(x) = 2e^x(x+1), f'(x) = \frac{e^x(x-1)}{x^2}, f'(x) = 6x\cos(3x^2+2), f'(x) = a^x \ln(a)$

Slide 3:

Slope = -12

Maxima @ $(x, y) = (-2, 20)$

Minima @ $(x, y) = (1, -7)$